Replace-One Sensitivity Proofs

All 49 Functions in the DPriv-Bench Category 1 Function Bank

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4 Tightness conclusions **18**

1 Setting and proof standard

Fix $n \geq 1$. A dataset is an ordered tuple

$$X = (x_1, \ldots, x_n) \in [0,1]^n := \mathcal{X}.$$

Datasets $X, X' \in \mathcal{X}$ are replace-one adjacent, written $X \sim X'$, when they differ in at most one coordinate. Since every query here is scalar, its global replace-one sensitivity is

$$\Delta_{\mathrm{ro}}(f) = \sup_{X \sim X'} |f(X) - f(X')|.$$

1.1 Tightness criterion

To state that a proposed upper bound s is tight (attainable), two checks are required:

$|f(X) - f(X')| \leq s$ for all $X \sim X'$ and all other parameters, eg: n, μ
$|f(X) - f(X')| = s$ for some $X \sim X'$ and some confuguration of other parameters

The second pair is called a tight witness. We use $\mathbf{0}^n = (0, \ldots, 0)$, $\mathbf{1}^n = (1, \ldots, 1)$, and e_k for the k-th standard basis vector. All logarithms are natural.

2 Reusable bounds

Lemma 1 (One-coordinate range)**.** *If $f(X) = \sum_{i=1}^n g_i(x_i)$, then*

$$\Delta_{\mathrm{ro}}(f) = \max_{1 \leq i \leq n} \left(\max_{t \in [0,1]} g_i(t) - \min_{t \in [0,1]} g_i(t) \right).$$

Proof. If coordinate k changes from a to b, every other summand cancels and

$$|f(X') - f(X)| = |g_k(b) - g_k(a)| \leq \max g_k - \min g_k.$$

Choose a coordinate with largest range and replace a minimizer by a maximizer to attain equality. □

Lemma 2 (Finite range)**.** *If $f([0,1]^n) \subseteq [A, B]$, then $\Delta_{\mathrm{ro}}(f) \leq B - A$. In particular, a binary query has sensitivity at most* 1*.*

Proof. For all adjacent X, X', both outputs lie in $[A, B]$, so their absolute difference is at most $B - A$. □

Lemma 3 (Open-chain local effect)**.** *For $f(X) = \sum_{i=1}^{n-1} h(x_i, x_{i+1})$, replacing coordinate k changes only*

$$d_k = \mathbf{1}[k > 1] + \mathbf{1}[k < n]$$

edges. If the change of each edge term is at most 1*, then*

$$|f(X') - f(X)| \leq \begin{cases} 0, & n = 1, \\ 1, & n = 2, \\ 2, & n \geq 3. \end{cases}$$

Proof. All unaffected edges cancel. Apply the triangle inequality to the remaining d_k terms and maximize d_k over k. □

Lemma 4 (Reverse triangle inequality)**.** *For vectors u, v, $\big|\|u\|_2 - \|v\|_2\big| \leq \|u - v\|_2$.*

3 Proofs for all functions

3.1 ID 1: $f(X) = \sum_{i=1}^{n} x_i$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. This is Lemma 1 with $g(t) = t$, whose range on $[0, 1]$ is 1. The adjacent pair $\mathbf{0}^n \to e_1$ changes the output from 0 to 1. □

3.2 ID 2: $f(X) = \prod_{i=1}^{n} x_i$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. If coordinate k changes from a to b, then

$$|f(X') - f(X)| = |b - a| \prod_{j \neq k} x_j \leq 1.$$

With all other coordinates equal to 1, the pair $\mathbf{1}^n - e_1 \to \mathbf{1}^n$ attains 1. □

3.3 ID 3: $f(X) = \max_{1 \leq i \leq n} x_i$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The output always lies in $[0, 1]$, so Lemma 2 gives the upper bound 1. The pair $\mathbf{0}^n \to e_1$ changes the maximum from 0 to 1. □

3.4 ID 4: $f(X) = \min_{1 \leq i \leq n} x_i$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The output is in $[0, 1]$. The adjacent pair $\mathbf{1}^n \to \mathbf{1}^n - e_1$ changes the minimum from 1 to 0, so the range bound is attained. □

3.5 ID 5: $f(X) = \frac{1}{n} \sum_{i=1}^{n} x_i$

Function-bank label: $\frac{1}{n}$. **Exact replace-one sensitivity:** $\frac{1}{n}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. Only one summand can change, by at most 1, so the mean changes by at most $1/n$. The pair $\mathbf{0}^n \to e_1$ attains $1/n$. □

3.6 ID 6: $f(X) = \sqrt{\sum_{i=1}^{n} x_i^2}$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. Write $f(X) = \|X\|_2$. If coordinate k changes from a to b, Lemma 4 gives

$$|f(X') - f(X)| \leq \|X' - X\|_2 = |b - a| \leq 1.$$

The pair $\mathbf{0}^n \to e_1$ attains 1. □

3.7 ID 7: $f(X) = \sum_{i=1}^{n-1} |x_i - x_{i+1}|$

Function-bank label: 2. **Exact replace-one sensitivity:** $\begin{cases} 0, & n = 1, \\ 1, & n = 2, \\ 2, & n \geq 3. \end{cases}$.

Tightness: The bank label 2 is tight exactly for $n \geq 3$.

Proof. Each edge term lies in $[0, 1]$, so Lemma 3 gives the displayed upper bound. For $n = 2$, $\mathbf{0}^2 \to e_1$ attains 1. For $n \geq 3$, $\mathbf{0}^n \to e_2$ creates two unit jumps and attains 2. For $n = 1$, the sum is empty and identically 0. □

3.8 ID 8: $f(X) = \sum_{i=1}^{n} |x_i - \bar{x}|, \quad \bar{x} = \frac{1}{n} \sum_{j=1}^{n} x_j$

Function-bank label: $\frac{2(n-1)}{n}$. **Exact replace-one sensitivity:** $\frac{2(n-1)}{n}$.

Tightness: The label is tight for every $n \geq 1$, including value 0 when $n = 1$.

Proof. Suppose coordinate k changes from a to b, and put $d = b - a$. Then $\bar{x}' = \bar{x} + d/n$. Each unchanged coordinate contributes a change at most $|d|/n$. For the changed coordinate,

$$(b - \bar{x}') - (a - \bar{x}) = \frac{n-1}{n} d,$$

so its absolute-deviation term changes by at most $(n-1)|d|/n$. Hence

$$|f(X') - f(X)| \leq \frac{2(n-1)}{n} |d| \leq \frac{2(n-1)}{n}.$$

For $\mathbf{0}^n \to e_1$, the new mean is $1/n$ and the new total absolute deviation is

$$\left|1 - \frac{1}{n}\right| + (n-1)\frac{1}{n} = \frac{2(n-1)}{n},$$

so the bound is exact. □

3.9 ID 9: $f(X) = \sum_{i=1}^{n}(x_i - \bar{x})^2$

Function-bank label: $\frac{n-1}{n}$. **Exact replace-one sensitivity:** $\frac{n-1}{n}$.

Tightness: The label is tight for every $n \geq 1$, including value 0 when $n = 1$.

Proof. For $n \geq 2$, fix the coordinates other than k, and set

$$S = \sum_{i \neq k} x_i, \qquad T = \sum_{i \neq k} x_i^2, \qquad r = \frac{S}{n-1} \in [0,1].$$

When $x_k = t$, the identity

$$f(t) = T + t^2 - \frac{(S+t)^2}{n} = C + \frac{n-1}{n}(t-r)^2$$

holds for a constant C. Its oscillation on $[0,1]$ is

$$\frac{n-1}{n}\max\{r^2, (1-r)^2\} \leq \frac{n-1}{n}.$$

The pair $\mathbf{0}^n \to e_1$ attains $(n-1)/n$. For $n = 1$, the query is identically 0. □

3.10 ID 10: $f(X) = \sum_{i=1}^{n}(x_i - \mu)^2, \quad \mu \in [0,1]$ fixed

Function-bank label: 1. **Exact replace-one sensitivity:** $\max\{\mu^2, (1-\mu)^2\}$.

Tightness: For a fixed query, the bank label 1 is tight only at $\mu = 0$ or $\mu = 1$. It is a tight uniform bound over the parameter family because $\sup_{\mu \in [0,1]} \Delta_{\mathrm{ro}}(f) = 1$.

Proof. Lemma 1 applies to $g_\mu(t) = (t-\mu)^2$. Its minimum is 0 at $t = \mu$, and its maximum is attained at the farther endpoint, giving

$$\Delta_{\mathrm{ro}}(f) = \max\{\mu^2, (1-\mu)^2\}.$$

Hold all other coordinates fixed and replace $x_k = \mu$ by 1 if $\mu \leq 1/2$, or by 0 if $\mu \geq 1/2$. This attains the stated maximum. For example, at $\mu = 1/2$ the exact sensitivity is $1/4$, not 1. □

3.11 ID 11: $f(X) = \sum_{i=1}^{n} x_i^2$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The single-coordinate function t^2 has range 1 on $[0,1]$. Lemma 1 gives the upper bound, and $\mathbf{0}^n \to e_1$ attains it. □

3.12 ID 12: $f(X) = \sum_{i=1}^{n} x_i^3$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The function t^3 increases from 0 to 1 on $[0, 1]$. Lemma 1 and the witness $\mathbf{0}^n \to e_1$ give the exact value 1. □

3.13 ID 13: $f(X) = \sum_{i=1}^{n} \sqrt{x_i}$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The function $\sqrt{t}$ has range 1 on $[0, 1]$. Lemma 1 applies, and $\mathbf{0}^n \to e_1$ is a tight witness. □

3.14 ID 14: $f(X) = \sum_{i=1}^{n} \log(1 + x_i)$

Function-bank label: $\log 2$. **Exact replace-one sensitivity:** $\log 2$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The summand increases from $\log 1 = 0$ to $\log 2$. Lemma 1 gives the bound and $\mathbf{0}^n \to e_1$ attains it. □

3.15 ID 15: $f(X) = \sum_{i=1}^{n} e^{x_i}$

Function-bank label: $e - 1$. **Exact replace-one sensitivity:** $e - 1$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The single-coordinate range of e^t on $[0, 1]$ is $e - 1$. Apply Lemma 1; replacing one zero by one attains the range. □

3.16 ID 16: $f(X) = \sum_{i=1}^{n} \sin(\pi x_i)$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. On $[0, 1]$, $\sin(\pi t) \in [0, 1]$, with values 0 at $t = 0$ and 1 at $t = 1/2$. Lemma 1 gives the upper bound; $\mathbf{0}^n \to \frac{1}{2} e_1$ attains it. □

3.17 ID 17: $f(X) = \sum_{i=1}^{n} \cos(\pi x_i)$

Function-bank label: 2. **Exact replace-one sensitivity:** 2.

Tightness: The label is tight for every $n \geq 1$.

Proof. On $[0, 1]$, $\cos(\pi t)$ ranges from 1 to -1, so its range is 2. Lemma 1 applies, and changing one coordinate from 0 to 1 attains 2. □

3.18 ID 18: $f(X) = \sum_{i=1}^{n} \arctan(x_i)$

Function-bank label: $\frac{\pi}{4}$. **Exact replace-one sensitivity:** $\frac{\pi}{4}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. $\arctan t$ increases from 0 to $\pi/4$ on $[0, 1]$. Lemma 1 and $\mathbf{0}^n \to e_1$ prove exactness. □

3.19 ID 19: $f(X) = \sum_{i=1}^{n} x_i(1 - x_i)$

Function-bank label: $\frac{1}{4}$. **Exact replace-one sensitivity:** $\frac{1}{4}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. $t(1 - t)$ has minimum 0 at the endpoints and maximum $1/4$ at $t = 1/2$. Lemma 1 gives the upper bound, and $\mathbf{0}^n \to \frac{1}{2}e_1$ attains it. □

3.20 ID 20: $f(X) = \sum_{i=1}^{n} x_i^2(1 - x_i)^2$

Function-bank label: $\frac{1}{16}$. **Exact replace-one sensitivity:** $\frac{1}{16}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The summand is $[t(1 - t)]^2$, so its range is $[0, (1/4)^2] = [0, 1/16]$. Apply Lemma 1; the pair $\mathbf{0}^n \to \frac{1}{2}e_1$ attains $1/16$. □

3.21 ID 21: $f(X) = \text{median}(x_1, \ldots, x_n)$

Function-bank label: 1.

Exact replace-one sensitivity: $\begin{cases} 1, & n \text{ odd}, \\ \frac{1}{2}, & n \text{ even and the two middle order statistics are averaged}, \\ 1, & n \text{ even and the lower or upper median is used}. \end{cases}$

Tightness: The bank label is always tight for odd n; for even n, it is tight for lower/upper median conventions but not for the averaged convention.

Proof. The median lies in $[0,1]$. For $n = 2m+1$,

$$(0^{m+1}, 1^m) \longrightarrow (0^m, 1^{m+1})$$

changes the median from 0 to 1, proving exact sensitivity 1.

For $n = 2m$ and the averaged convention, fix the other $2m-1$ ordered observations $y_1 \le \cdots \le y_{2m-1}$, set $y_0 = 0$ and $y_{2m} = 1$, and change the inserted value from 0 to 1. Monotonicity of the median in the inserted value shows this gives the largest possible change, namely

$$\frac{1}{2}\,(y_{m+1} - y_{m-1}) \le \frac{1}{2}.$$

The pair

$$(0^{m+1}, 1^{m-1}) \longrightarrow (0^m, 1^m)$$

attains $1/2$. For the lower median, use $(0^m, 1^m) \to (0^{m-1}, 1^{m+1})$; for the upper median, use the averaged-convention witness above. Each changes the selected order statistic from 0 to 1. □

3.22 ID 22: $f(X) = Q_3 - Q_1$ (interquartile range)

Function-bank label: 1.

Nearest-rank convention. Let $Q_p = x_{(\lceil np \rceil)}$. Then

$$\textbf{Exact replace-one sensitivity:} \begin{cases} 0, & n = 1, \\ 1, & n \ge 2. \end{cases}.$$

Tightness: The bank label is tight for $n \ge 2$, and is not tight for $n = 1$.

Proof. The IQR lies in $[0,1]$. For $n \ge 2$, put $r = \lceil 3n/4 \rceil$ and use

$$X = (0^r, 1^{n-r}) \longrightarrow X' = (0^{r-1}, 1^{n-r+1}).$$

Because $\lceil n/4 \rceil < r$, Q_1 remains 0, while Q_3 changes from 0 to 1. Thus the range bound is attained. For $n = 1$, both quartiles are the only observation and the IQR is zero. □

Type-7 linear interpolation. For

$$h_p = (n-1)p, \quad j_p = \lfloor h_p \rfloor, \quad \gamma_p = h_p - j_p,$$

define

$$Q_p^{(7)} = \begin{cases} x_{(j_p+1)}, & \gamma_p = 0, \\ (1-\gamma_p)x_{(j_p+1)} + \gamma_p x_{(j_p+2)}, & 0 < \gamma_p < 1. \end{cases}$$

Then

$$\textbf{Exact replace-one sensitivity:} \begin{cases} 0, & n = 1, \\ \frac{1}{2}, & n = 2, \\ 1, & n \ge 3,\ n \equiv 1 \pmod 4, \\ \frac{3}{4}, & n \ge 3,\ n \equiv 0 \text{ or } 2 \pmod 4, \\ \frac{1}{2}, & n \ge 3,\ n \equiv 3 \pmod 4. \end{cases}.$$

Tightness: Under type-7 interpolation, the bank label 1 is tight exactly when $n \ge 5$ and $n \equiv 1 \pmod 4$.

Proof. The IQR is an L-statistic. For every $L(X) = \sum_i c_i x_{(i)}$, the layer-cake identity

$$L(X) = \int_0^1 L\big(\mathbf{1}[x_1 > s], \ldots, \mathbf{1}[x_n > s]\big)\, ds$$

shows that continuous-domain sensitivity is at most binary-domain sensitivity. The reverse inequality holds because binary datasets are a subset of $[0,1]^n$, so the two sensitivities are equal.

Let $B_k = (0^{n-k}, 1^k)$, $a = (n-1)/4$, and $c(u) = \min\{\max\{u, 0\}, 1\}$. Direct substitution of binary order statistics into the type-7 formula gives

$$\mathrm{IQR}^{(7)}(B_k) = c(k-a) - c(k-3a).$$

A binary replacement changes k by one, hence

$$\Delta_{\mathrm{ro}}(f) = \max_{0 \le k < n} \left| \mathrm{IQR}^{(7)}(B_{k+1}) - \mathrm{IQR}^{(7)}(B_k) \right|.$$

Evaluating the two clipped-linear terms for the four residue classes of n mod 4 gives the displayed piecewise values. Equality is attained by $B_k \to B_{k+1}$, with $k = 0$ when $n = 2$, and $k = m$ for $n = 4m, 4m+1, 4m+2$, or $4m+3$. □

3.23 ID 23: $f(X) = \#\{i : x_i > 0.5\}$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \ge 1$.

Proof. Replacing one record changes only one indicator, so the count changes by at most 1. The pair $\mathbf{0}^n \to e_1$ flips exactly one indicator and attains the bound. □

3.24 ID 24: $f(X) = \#\{i : x_i \in [a, b]\}$

Function-bank label: 1.

Let $A = [a, b] \cap [0, 1]$. Then

$$\textbf{Exact replace-one sensitivity:} \begin{cases} 1, & A \neq \varnothing \text{ and } [0,1] \setminus [a,b] \neq \varnothing, \\ 0, & \text{otherwise.} \end{cases}.$$

Tightness: The label is tight exactly when the membership indicator is nonconstant on the data domain.

Proof. Only one membership indicator can change, so the sensitivity is at most 1. If both an inside point $u_{\mathrm{in}} \in A$ and an outside point $u_{\mathrm{out}} \in [0,1] \setminus [a,b]$ exist, replace one by the other to change the count by 1. If either set is empty, the indicator is constant and the exact sensitivity is 0. □

3.25 ID 25: $f(X) = \sum_{i=1}^{n-1} x_i x_{i+1}$

Function-bank label: 2. **Exact replace-one sensitivity:** $\begin{cases} 0, & n = 1, \\ 1, & n = 2, \\ 2, & n \ge 3. \end{cases}$.

Tightness: The bank label 2 is tight exactly for $n \ge 3$.

Proof. Every edge product lies in $[0,1]$, so Lemma 3 gives the piecewise upper bound. For $n=2$, $(0,1)\to(1,1)$ attains 1. For $n\ge 3$, $\mathbf{1}^n - e_2 \to \mathbf{1}^n$ changes two adjacent products from 0 to 1, attaining 2. For $n=1$, the sum is empty. □

3.26 ID 26: $f(X)=\sum_{1\le i<j\le n} x_i x_j$

Function-bank label: $n-1$. **Exact replace-one sensitivity:** $n-1$.

Tightness: The label is tight for every $n\ge 1$, with value 0 at $n=1$.

Proof. If coordinate k changes from a to b, then

$$|f(X')-f(X)| = |b-a|\sum_{j\neq k} x_j \le n-1.$$

The pair $\mathbf{1}^n - e_1 \to \mathbf{1}^n$ attains $n-1$. □

3.27 ID 27: $f(X)=\sum_{i=1}^{\lfloor n/2\rfloor}(x_{2i-1}-x_{2i})$

Function-bank label: 1. **Exact replace-one sensitivity:** $\begin{cases}0, & n=1,\\ 1, & n\ge 2.\end{cases}$

Tightness: The bank label is tight exactly for $n\ge 2$.

Proof. For $n\ge 2$, every included coordinate has coefficient $+1$ or -1, so replacing one value changes the sum by at most 1. The pair $\mathbf{0}^n \to e_1$ attains 1. For $n=1$, the sum is empty. □

3.28 ID 28: $f(X)=\left(\sum_{i=1}^{n} x_i\right)^2$

Function-bank label: $2n-1$. **Exact replace-one sensitivity:** $2n-1$.

Tightness: The label is tight for every $n\ge 1$.

Proof. Let $s=\sum_{j\neq k} x_j \in [0,n-1]$. Since the square is increasing on the nonnegative line, the largest change as x_k varies over $[0,1]$ is

$$(s+1)^2 - s^2 = 2s+1 \le 2n-1.$$

Taking $s=n-1$, equivalently $\mathbf{1}^n - e_1 \to \mathbf{1}^n$, attains equality. □

3.29 ID 29: $f(X)=\sum_{i=1}^{n} i\,x_i$

Function-bank label: n. **Exact replace-one sensitivity:** n.

Tightness: The label is tight for every $n\ge 1$.

Proof. At coordinate k, the change is $k|b-a|\le k\le n$. Changing the last coordinate from 0 to 1, $\mathbf{0}^n \to e_n$, attains n. □

3.30 ID 30: $f(X) = \mathbf{1}\left[\sum_{i=1}^{n} x_i > \frac{n}{2}\right]$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. The output is binary, so Lemma 2 gives an upper bound of 1. Starting from $X = (1/2)^n$, replace $x_1 = 1/2$ by 1. The strict-threshold output changes from 0 to 1. □

3.31 ID 31: $f(X) = \sum_{i=1}^{n} \mathbf{1}[x_i > 0.7]$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. Only one indicator can change, so the sum changes by at most 1. The pair $\mathbf{0}^n \to e_1$ flips one indicator and attains 1. □

3.32 ID 32: $f(X) = \prod_{i=1}^{n} \mathbf{1}[x_i > \tau]$

Function-bank label: 1. **Exact replace-one sensitivity:** $\begin{cases} 1, & 0 \leq \tau < 1, \\ 0, & \tau < 0 \text{ or } \tau \geq 1. \end{cases}$.

Tightness: The bank label is tight exactly when $0 \leq \tau < 1$.

Proof. The output is binary. If $0 \leq \tau < 1$, $\mathbf{1}^n - e_1 \to \mathbf{1}^n$ changes the product from 0 to 1. If $\tau < 0$, every factor is always 1; if $\tau \geq 1$, every factor is always 0 because the inequality is strict. The query is constant in either case. □

3.33 ID 33: $f(X) = \mathbf{1}[x_1 > x_2]$

Function-bank label: 1. **Exact replace-one sensitivity:** 1 $(n \geq 2)$.

Tightness: For every admissible $n \geq 2$, the label is tight.

Proof. The output is binary. The adjacent pair

$$(0, \tfrac{1}{2}, 0, \ldots, 0) \longrightarrow (1, \tfrac{1}{2}, 0, \ldots, 0)$$

changes it from 0 to 1, attaining the binary range bound. □

3.34 ID 34: $f(X) = \left(\sum_{i=1}^{n} \mathbf{1}[x_i > 0.5]\right) \bmod 2$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. Parity is binary, so its change is at most 1. The pair $\mathbf{0}^n \to e_1$ flips one indicator and hence flips parity, attaining 1. □

3.35 ID 35: $f(X) = \log\left(1 + \sum_{i=1}^{n} x_i^2\right)$

Function-bank label: $\log 2$. **Exact replace-one sensitivity:** $\log 2$.

Tightness: The label is tight for every $n \geq 1$.

Proof. Fix the other coordinates and let $c = \sum_{i \neq k} x_i^2 \geq 0$. As $x_k = t$ ranges over $[0, 1]$, the oscillation is

$$\log(1 + c + 1) - \log(1 + c) = \log \frac{c+2}{c+1} \leq \log 2.$$

The maximum occurs at $c = 0$; $\mathbf{0}^n \to e_1$ attains it. □

3.36 ID 36: $f(X) = \sqrt{\sum_{i=1}^{n} (x_i - x_{i+1 \bmod n})^2}$

Function-bank label: $\sqrt{2}$. **Exact replace-one sensitivity:** $\begin{cases} 0, & n = 1, \\ \sqrt{2}, & n \geq 2. \end{cases}$

Tightness: The bank label is tight exactly for $n \geq 2$.

Proof. Let $A(X) = (x_1 - x_2, \ldots, x_n - x_1)$, so $f(X) = \|A(X)\|_2$. For $n \geq 2$, changing one coordinate by d changes exactly two cyclic differences, by d and $-d$. Therefore

$$|f(X') - f(X)| \leq \|A(X') - A(X)\|_2 = \sqrt{2}\,|d| \leq \sqrt{2}.$$

The pair $\mathbf{0}^n \to e_1$ attains $\sqrt{2}$, including at $n = 2$, where both directed cyclic edges appear. For $n = 1$, the only difference is $x_1 - x_1 = 0$. □

3.37 ID 37: $f(X) = \dfrac{\sum_{i=1}^{n} x_i^2}{1 + \sum_{i=1}^{n} x_i}$

Function-bank label: $\frac{1}{2}$. **Exact replace-one sensitivity:** $\frac{1}{2}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. Fix all coordinates except k, and write

$$s = \sum_{i \neq k} x_i, \qquad q = \sum_{i \neq k} x_i^2, \qquad c = 1 + s.$$

Then $c \geq 1$, $0 \leq q \leq s = c - 1$, and, for $x_k = t$,

$$h(t) = \frac{q + t^2}{c + t}, \quad h'(t) = \frac{t^2 + 2ct - q}{(c+t)^2}, \quad h''(t) = \frac{2(c^2 + q)}{(c+t)^3} > 0.$$

Thus h is strictly convex. Its minimum occurs at

$$r = \sqrt{c^2 + q} - c, \qquad r^2 + 2cr = q, \qquad 0 \leq r < 1.$$

Also

$$h(1) - h(0) = \frac{c - q}{c(c+1)} > 0,$$

so the larger endpoint is $t = 1$. Substituting the relation for r gives

$$\operatorname{osc}_{[0,1]} h = h(1) - h(r) = \frac{(1-r)^2}{c+1} \leq \frac{1}{2}.$$

When all other coordinates are zero, $c = 1, q = r = 0$; $\mathbf{0}^n \to e_1$ changes the query from 0 to 1/2. □

3.38 ID 38: $f(X) = \exp\left(-\sum_{i=1}^{n} x_i^2\right)$

Function-bank label: $1 - e^{-1}$. **Exact replace-one sensitivity:** $1 - e^{-1}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. Fix $c = \sum_{i \neq k} x_i^2 \geq 0$. The coordinate-wise oscillation is

$$e^{-c} - e^{-c-1} = e^{-c}(1 - e^{-1}) \leq 1 - e^{-1}.$$

The pair $\mathbf{0}^n \to e_1$ has $c = 0$ and attains equality. □

3.39 ID 39: $f(X) = \sum_{i=1}^{n} \frac{1}{1 + x_i^2}$

Function-bank label: $\frac{1}{2}$. **Exact replace-one sensitivity:** $\frac{1}{2}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The summand decreases from 1 at $t = 0$ to $1/2$ at $t = 1$, so its range is $1/2$. Lemma 1 and $\mathbf{0}^n \to e_1$ prove exactness. □

3.40 ID 40: $f(X) = \sum_{i=1}^{n} \sigma(x_i), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$

Function-bank label: $\frac{1-e^{-1}}{2(1+e^{-1})}$. **Exact replace-one sensitivity:** $\frac{1-e^{-1}}{2(1+e^{-1})}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The sigmoid is increasing, so its range on $[0, 1]$ is

$$\sigma(1) - \sigma(0) = \frac{e}{e+1} - \frac{1}{2} = \frac{e-1}{2(e+1)} = \frac{1 - e^{-1}}{2(1 + e^{-1})}.$$

Apply Lemma 1; $\mathbf{0}^n \to e_1$ attains this value. □

3.41 ID 41: $f(X) = \frac{\sum_{i=1}^{n} x_i e^{x_i}}{\sum_{j=1}^{n} e^{x_j}}$

Function-bank label: $\frac{e}{e+n-1}$. **Exact replace-one sensitivity:** $\frac{e}{e+n-1}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. Fix the other coordinates and put

$$A = \sum_{i \neq k} e^{x_i}, \qquad B = \sum_{i \neq k} x_i e^{x_i}.$$

Because $x_i \in [0, 1]$, $A \geq n - 1$ and $0 \leq B \leq A$. For $x_k = t$,

$$h(t) = \frac{B + te^t}{A + e^t}, \qquad h'(t) = \frac{e^t[A(1 + t) - B + e^t]}{(A + e^t)^2} > 0.$$

Thus the greatest single-coordinate change is $h(1) - h(0)$, and

$$h(1) - h(0) = \frac{e(A+1) - (e-1)B}{(A+e)(A+1)} \leq \frac{e}{A+e} \leq \frac{e}{n-1+e}.$$

For $\mathbf{0}^n \to e_1$, $A = n-1$ and $B = 0$, so every inequality is an equality. □

3.42 ID 42: $f(X) = \sum_{i=1}^{n} \left| x_i - \frac{1}{2} \right|$

Function-bank label: $\frac{1}{2}$. **Exact replace-one sensitivity:** $\frac{1}{2}$.

Tightness: The label is tight for every $n \geq 1$.

Proof. The summand ranges from 0 at $t = 1/2$ to $1/2$ at either endpoint. Lemma 1 gives the exact value; for example, $\mathbf{0}^n \to \frac{1}{2}e_1$ changes the sum by $1/2$. □

3.43 ID 43: $f(X) = \sum_{i=1}^{n-1} (x_i - x_{i+1})^2$

Function-bank label: 2. **Exact replace-one sensitivity:** $\begin{cases} 0, & n = 1, \\ 1, & n = 2, \\ 2, & n \geq 3. \end{cases}$.

Tightness: The bank label 2 is tight exactly for $n \geq 3$.

Proof. Each squared edge difference lies in $[0, 1]$, so Lemma 3 gives the displayed upper bound. $\mathbf{0}^2 \to e_1$ attains 1, while $\mathbf{0}^n \to e_2$ creates two unit squared differences and attains 2 for $n \geq 3$. The $n = 1$ sum is empty. □

3.44 ID 44: $f(X) = \sum_{i=1}^{n} x_{\pi(i)}$ (π **fixed**)

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \geq 1$.

Proof. A fixed permutation reorders the terms but does not change their sum: $\sum_i x_{\pi(i)} = \sum_i x_i$. The proof and witness are therefore those of ID 1. □

3.45 ID 45: $f(X) = \mathrm{rank}(x_k)$

Function-bank label: $n - 1$. **Exact replace-one sensitivity:** $n - 1$.

Tightness: Under any common ties convention satisfying that range, the label is tight for every $n \geq 1$, with value 0 at $n = 1$.

Proof. The rank lies in $[1, n]$, so its change is at most $n - 1$. Choose the other $n - 1$ coordinates to be distinct interior values, for example $1/n, 2/n, \ldots, (n-1)/n$, and change the target coordinate x_k from 0 to 1. There are no ties, and its rank changes from 1 to n. Thus the range bound is attained regardless of how ties elsewhere would be resolved. □

3.46 ID 46: $f(X) = \sum_{\substack{1 \le i \le n \\ i \text{ even}}} x_i$

Function-bank label: 1. **Exact replace-one sensitivity:** $\begin{cases} 0, & n = 1, \\ 1, & n \ge 2. \end{cases}$

Tightness: The bank label is tight exactly for $n \ge 2$.

Proof. For $n \ge 2$, replacing an even-position coordinate changes the sum by at most 1, and $\mathbf{0}^n \to e_2$ attains 1. When $n = 1$, there is no even position and the sum is empty. □

3.47 ID 47: $f(X) = \sum_{i=1}^{n} \frac{x_i}{i}$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \ge 1$.

Proof. At position i, a replacement changes the query by at most $1/i$. The largest coefficient is 1, at $i = 1$, and $\mathbf{0}^n \to e_1$ attains it. □

3.48 ID 48: $f(X) = -\sum_{i=1}^{n} x_i \log(x_i + \delta), \quad \delta > 1$

Function-bank label: $\log(1 + \delta)$. **Exact replace-one sensitivity:** $\log(1 + \delta)$.

Tightness: The label is tight for every $n \ge 1$ and every fixed $\delta > 1$.

Proof. For $g_\delta(t) = -t \log(t + \delta)$,

$$g'_\delta(t) = -\log(t + \delta) - \frac{t}{t + \delta} < 0$$

because $t \in [0, 1]$ and $\delta > 1$. Hence g_δ decreases from $g_\delta(0) = 0$ to $g_\delta(1) = -\log(1 + \delta)$, and its range is $\log(1 + \delta)$. Lemma 1 applies, with tight witness $\mathbf{0}^n \to e_1$. □

3.49 ID 49: $f(X) = \sum_{i=1}^{n} \left| x_i - \frac{i}{n} \right|$

Function-bank label: 1. **Exact replace-one sensitivity:** 1.

Tightness: The label is tight for every $n \ge 1$.

Proof. For $c_i = i/n \in [0, 1]$, the range of $|t - c_i|$ over $t \in [0, 1]$ is

$$\max\{c_i, 1 - c_i\} \le 1.$$

Lemma 1 gives the upper bound. At $i = n$, $c_n = 1$, and changing x_n from 0 to 1, namely $\mathbf{0}^n \to e_n$, changes that term from 1 to 0, attaining 1. □

4 Tightness conclusions

Under $X \in [0,1]^n$, fixed n, and replace-one adjacency, each query has an exact upper bound and a matching single-coordinate witness once the necessary parameter and convention choices are made. The raw function-bank label should not be read as unconditionally tight (as contradict to the tightness we considered in section 1.1) in the following cases:

- IDs 7, 25, and 43 require $n \geq 3$ for the label 2.
- ID 10 has fixed-μ sensitivity $\max\{\mu^2, (1-\mu)^2\}$; the label 1 is attained at $\mu \in \{0, 1\}$.
- ID 21 depends on the even-sample median convention.
- ID 22 depends on the quantile convention and on n mod 4 under type-7 interpolation.
- IDs 24 and 32 require nondegenerate interval/threshold parameters.
- IDs 27, 36, and 46 have sensitivity 0 at $n = 1$.
- ID 33 requires $n \geq 2$, and ID 45 requires a rank convention.